

Coding and Smart Systems Outreach Program

Program Lead: Dr. Nitindra Chowdary (Niti) NPAVULURI@RSU.EDU

Pre-program set-up instructions (Program Lead):

- Assemble and boot each Pi. A window will be opened to create an admin account with a username and password.
- Create an admin account.
- Connect to a Wi-Fi network if available to update OS.
- Log in using admin account and click the terminal icon (a black monitor box) in the top-left menu bar to open a terminal. Keep this terminal open.
- Update OS packages. Press enter key after typing each of the following commands:
 - `sudo apt update`
 - `sudo apt full-upgrade -y`
- Install required libraries. The following libraries are installed for projects listed in this program (in the same terminal press enter after typing each command below):
 - `sudo apt install flac`
 - `sudo apt install libcap-dev`
- Check if Python is installed. If not, install it:
 - Check if it is installed: `python3 --version`
 - If not, install it: `sudo apt install python3`
- Close the terminal.
- Right click on the desktop screen, select new folder to create a new folder and name it as projects. You can copy open-source projects or your custom projects into this folder.
- Double click on the projects folder to open it and create a folder named venv inside the projects folder. Python libraries will be installed in this folder.
- Open the top left menu bar, click on the programming menu and click on Thonny to open it. Thonny is used for running Python programs.
- In Thonny, click on Run, click on configure interpreter (a pop-up window opens) and then click on New virtual environment in the pop-up window. This is recommended because any python libraries installed for this program will be contained in this environment.
- Another pop-up window opens that says you need to select an empty folder, click on OK, browse and select the venv folder, and click on select folder.

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- After the progress bar is completed, click ok.
- Click on File, then click on New
- Type the following code in the window: `print("Hello")`
- Click on File, then click on save as, a new window should open.
- Browse and select the projects folder on the Desktop.
- Type a file name: test and then click save.
- Click on Run and then click on Run current script to run the test.py file.
- You should be able to see the output Hello in the Shell window at the bottom.
- You can install python libraries directly in the shell window using pip command:
 - `!pip install <package_name>`
 - Example: `!pip install RPLCD` (library to display text on LCD screen)
- To run the open-source Raspberry Pi projects:
 - Power off the Pi
 - Connect the required hardware components,
 - Make sure all wiring connections are correct
 - Power on the Pi
 - Log in using admin or other account
 - Open Thonny application
 - Click File, then click on open, browse the project you would like to run
 - Click on the .py file to open it in Thonny
 - Run the file
 - You can see the text output in the Shell window and the visual output on the hardware components
- Instead of using admin account, you can create student accounts:
 - Log in to Pi using admin account
 - Open a Terminal
 - Type the following command and press enter: `sudo adduser <username>`
 - Example: `sudo adduser student`
 - Follow the prompts and enter required information to create the student account.
 - **Note:** You may not see the password you type. This is a security feature. So, just type the password correctly and press enter
 - Log out of the admin account and log in using student account

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- You need to enable the LCD display to use LCD:
 - Log in using the admin account
 - Open a terminal
 - Type the following command and press enter: `sudo raspi-config`
 - A new window should open
 - Click on Interface options (you may need to use arrow keys to navigate options and press enter key)
 - Click on Enable I2C.

Troubleshooting Tips:

- **Hardware:**
 - If LED does not work, resistor may be bad or it is connected to one GPIO pin number and a different number is used in the Python program.
 - Jumper wire tips must be straight and not bent.
 - GPIO extension board must be completely seated on the breadboard.
 - GPIO pin numbers of hardware components should always match pin numbers defined in the Python program.
- **Software:**
 - Python program not running or displaying errors:
 - Check indentation (spacing before lines must be uniform)
 - Check [syntax rules](#)
 - Missing libraries (install them using `!pip` command in shell window)
 - Libraries needed for Pi OS that are installed using the `sudo` command in a terminal must be installed after logging in with an admin account.
 - If using multiple workstations in a lab setting, configure the same Thonny theme on all computers for quick error identification and easier troubleshooting:
 - Open Thonny
 - Click on Tools menu option on the top
 - Click on Options
 - Click on Theme & Font
 - Select UI theme: Windows
 - Select Syntax theme: IDLE Classic